

Optimizing What Is Verified: Binarization-Aware Latent Refinement for Generative Inverse Design of Auxetic Metamaterials

Trình Bình Minh*

Draft, October 8, 2026

Abstract

Inverse design of auxetic metamaterials asks for a unit cell whose verified effective properties match a prescribed pair of Poisson ratios (ν_{12}, ν_{21}) . Topology optimization solves each target separately with hundreds of finite-element (FE) solves; generative models return candidates almost instantly but rarely meet tight tolerances once the binarized design is analysed. We study a pipeline that samples a conditional variational autoencoder (cVAE) and refines its latent code with exact homogenization sensitivities, and show why naive refinement fails: it drives the objective to $\sim 10^{-10}$ on grey density fields while the FE-verified error stays six orders of magnitude larger. Optimizing what is verified, by placing Heaviside projection and the exact resampling inside the objective and selecting by verified error, raises the verified R^2 of ν_{12} of two-dimensional cells from 0.874 to 0.999 for single samples (0.831 to 0.989 on a second target set) and lowers the error after best-of-30 selection by 59–68% on three disjoint sets. Against topology optimization from scratch, verified by the same chain, the pipeline reaches the same accuracy with about eight times fewer FE solves and, in a pre-specified test on targets of moderate anisotropy, a 27% lower joint error of (ν_{12}, ν_{21}) (95% CI 8–42%). Adding corner-contact and thin-feature penalties to the same objective raises the manufacturable fraction from 0.31 to 0.72–0.76, close to topology optimization, at the cost of this joint-error advantage. Topology optimization remains the better choice for strongly anisotropic targets and outside the training range, and we state when each method should be used. Topology optimization with Heaviside continuation shows the same objective–verification gap, and both methods occasionally return degenerate cells, so verification must include a degeneracy check.

Keywords: Auxetic metamaterials; Inverse homogenization; Topology optimization; Deep generative models; Heaviside projection; Verification

1 Introduction

Auxetic metamaterials contract laterally under uniaxial compression. Their negative Poisson’s ratio originates in the microarchitecture rather than the chemistry of the constituent material [1–3]. Designing them is an inverse problem: given a target pair of Poisson ratios (ν_{12}^*, ν_{21}^*) , find a unit cell whose effective properties, verified by finite-element (FE) analysis of the manufacturable design, match it. Inverse homogenization by topology optimization is the classical solution [4–6], but every new target requires an iterative optimization with hundreds of FE solves. Data-driven inverse design amortizes this cost: variational autoencoders, generative adversarial networks and diffusion models trained on simulated libraries return candidate architectures almost instantly [7–14]. Neural networks also enter topology optimization directly, as reparameterizations of the density field [15, 16]; a critical review of these approaches found that many reported gains disappear when they are compared with well-tuned classical optimization [17]. Whether a generative model is a practical solver for auxetic inverse design therefore depends on two questions: can it reach the verified accuracy of topology optimization, and for which targets?

*[Affiliation and corresponding-author details to be inserted.]

What is manufactured is a binarized, resampled version of a density field, so accuracy must be judged by FE analysis of that design rather than by a learned surrogate or by the continuous field. Surrogates of the forward map are known to be exploitable by optimizers and generators [18], and single samples rarely meet tight tolerances, so pipelines generate many candidates and select among them with FE [13]. A natural next step is to refine the selected design with gradients of an exact solver. We show that this step fails in an instructive way: the optimizer exploits intermediate densities that do not survive binarization, exactly as it would exploit an inaccurate surrogate. The same happens in topology optimization with Heaviside continuation. Closing this gap is what turns the generative pipeline into an accurate inverse-design solver.

This paper makes the following contributions.

1. **A verified inverse-design pipeline for auxetic cells.** Best-of-30 sampling from a cVAE followed by guarded latent refinement returns cells whose FE-verified ν_{12} reaches $R^2 = 0.995\text{--}0.998$ on three disjoint sets of 100 held-out targets, with about 107 FE solves per target.
2. **Optimizing what is verified.** We give direct evidence that gradient-based search over density fields, in the latent space of a generative model as well as over element densities in topology optimization, reaches near-zero objectives on fields whose verified error is orders of magnitude larger. Placing differentiable counterparts of the verification operators (Heaviside projection with continuation [19, 20] and the exact resampling) inside the objective, and selecting by verified error, closes most of the gap. The improvement is replicated on two further disjoint target sets.
3. **A like-for-like comparison with topology optimization from scratch.** Both methods are verified by the same chain and compared with paired confidence intervals, stratified by target anisotropy, including a pre-specified confirmation on a fresh set and a pre-specified hybrid that failed.
4. **When to use which method.** We report where the generative pipeline loses: strongly anisotropic targets, manufacturability unless it is penalized during refinement (which then costs accuracy), and targets with an unseen sign of the Poisson ratio, where it beats retrieval from the training library but not topology optimization. Together with the cost of the training data, these results give a practical rule for choosing between the two methods (Table 7). Controlled ablations of a Kolmogorov–Arnold network (KAN) head and an implicit wavelet (WIRE) decoder are negative.

Every reported accuracy is FE-verified on the binarized design, every comparison carries a paired bootstrap confidence interval, and success criteria were written down before each experiment (Section 2.8).

2 Methods

2.1 Data generation

Unit cells were generated by SIMP topology optimization [21–23] with a density filter [24], periodic boundary conditions and energy-based homogenization [25–27]. Auxetic objectives were minimized from four seed topologies (hourglass, hexagonal, re-entrant bow-tie, circular inclusion), using the optimality-criteria update or the method of moving asymptotes [28] depending on the seed. Process parameters (volume fraction, penalization, filter radius, move limit, void size, lattice rotation) were screened by Latin hypercube sampling [29] and refined in an adaptive multi-batch design of experiments using Sobol’ sequences [30] and optimized Latin hypercubes. Samples with oscillatory convergence, degenerate topologies or labels outside the physical range were removed. The 50×50 density fields were box-filtered to 64×64 images and labelled with the homogenized (ν_{12}, ν_{21}) . The production dataset contains 13 624 cells. They were split 70/15/15 at random, stratified by seed topology and by five fixed bins of ν_{12} , so that every split covers all seeds and the full ν_{12} range (2 044 test cells). Only after splitting were the training images augmented sixfold by rotations and reflections, with $\nu_{12} \leftrightarrow \nu_{21}$ swapped under 90° rotations, giving 57 216 training samples; no augmented copy of a training cell can therefore appear in the validation or test split. Generating the dataset required an estimated $2.7\text{--}3.0 \times 10^6$ FE solves (65–72 CPU hours).

2.2 Homogenization and exact sensitivities

Density fields $\rho \in [0, 1]^{50 \times 50}$ are discretized with bilinear quadrilateral elements and the modified SIMP interpolation $E(\rho) = E_{\min} + \rho^p(E_0 - E_{\min})$ with $E_{\min} = 10^{-9}E_0$, $p = 3$ and base-material Poisson ratio $\nu_0 = 0.3$. The homogenized stiffness $\mathbf{Q}$ follows from three periodic load cases through element mutual energies [27]. Poisson ratios are obtained from $\mathbf{S} = \mathbf{Q}^{-1}$ as $\nu_{12} = -S_{12}/S_{11}$ and $\nu_{21} = -S_{12}/S_{22}$, which remains valid for rotated, fully anisotropic cells. Because the energy formulation is self-adjoint, the sensitivities $\partial \mathbf{Q} / \partial \rho_e$ are available without extra solves. They are propagated analytically through $\partial \mathbf{S} = -\mathbf{S}(\partial \mathbf{Q})\mathbf{S}$ and the quotient rule in a custom PyTorch [31] autograd function, whose gradients agree with finite differences to a relative error below 10^{-4} . The solver agrees with an independent scikit-fem [32] implementation to 10^{-9} on 24 designs. One solve with sensitivities takes 0.086 s.

2.3 Conditional VAE

The cVAE [33, 34] is conditioned on $c = (\nu_{12}, \nu_{21})$ and has a 32-dimensional latent space. The encoder consists of four convolution–batch-norm–ReLU–pooling blocks (32 to 256 channels) that keep a 4×4 spatial map, followed by linear layers for the mean and log-variance. The decoder mirrors the encoder with transposed convolutions and a sigmoid output. Stage 1 trains with binary cross-entropy reconstruction, a KL term with linear warm-up [35] and a property-consistency term from a frozen convolutional surrogate (weight $\gamma = 20$; surrogate test R^2 of 0.974 for ν_{12} and 0.964 for ν_{21}). Stage 2 resumes from stage 1 and adds the squared error between c and the FE-homogenized Poisson ratios of decoded *prior* samples $z \sim \mathcal{N}(0, I)$ (weight 20; 8 samples every second epoch). This is the regime used at inference, and training with this loss from scratch fails. Optimization uses Adam [36]. Checkpoints are selected by FE-verified R^2 on validation conditions rather than by validation loss, which the KL warm-up makes unreliable.

2.4 Verification, degeneracy and manufacturability

A decoded image is verified by (1) thresholding at 0.5; (2) nearest-neighbour resampling to 50×50 with the index map of the imaging library used to build the dataset; and (3) FE homogenization. Periodic boundary conditions on the element mesh make every pixel cell tile the plane by translation, so no matching of opposite edges is required. An earlier version of the pipeline averaged opposite boundary rows and columns before thresholding; we removed this step because it altered boundary pixels without being needed for tiling (Appendix A). A verified design is called *degenerate* if $\min(E_1, E_2) < 10^{-3}E_0$, i.e. it has no load path in at least one direction; such a cell is counted as a failure in all analyses but kept in all averages.

A design is called manufacturable if its solid phase is a single connected component (8-connectivity) and no component vanishes after morphological erosion by one pixel, i.e. no strut is thinner than two pixels. For a cell of side L this is a minimum feature of $0.04L$, e.g. 0.4 mm for a 10 mm cell, comparable to the nozzle diameter of desktop fused-filament printers. Because 8-connectivity admits corner-only contacts (one-node hinges), we checked their prevalence in the training data: 17% of the images contain one, but the share is 28% among cells with small $|\nu_{12}|$ and 3% among cells with large $|\nu_{12}|$, so the extreme Poisson ratios in our data do not originate from hinges. Among final designs, 16–26% of the generated cells and 7–13% of those from topology optimization contain such a contact; subdividing pixels for the mesh-convergence check keeps a corner contact a single node, so that check does not detect them. We therefore also report manufacturability with 4-connectivity, under which corner-only contacts do not connect.

Among the candidates for a target, the pipeline selects either by verified joint error $|\Delta \nu_{12}| + |\Delta \nu_{21}|$ (Sections 3.2–3.3) or by a composite score $0.6 s_{\text{acc}} + 0.3 s_{\text{man}} + 0.1 s_{\text{aes}}$ (Section 3.5). The accuracy score s_{acc} is $|\Delta \nu_{12}|$ min–max normalized within the candidate pool, with sign reversed; s_{man} is the mean of the connectivity and feature-size indicators; the aesthetic score s_{aes} averages mirror symmetry about both axes and an inverse normalized perimeter-to-area ratio with periodic wrap-around. The weights encode the priority order and are a design choice, not a learned quantity.

2.5 Latent refinement

Given ν^* and a starting latent z_0 (single sample or best-of- N winner), binarization-aware refinement minimizes

$$\mathcal{L}_\beta(z) = \frac{1}{2} \|\nu(\mathcal{R} \circ H_\beta \circ D(z, c)) - \nu^*\|_2^2, \quad (1)$$

where D is the frozen decoder, $\mathcal{R}$ the nearest-neighbour resampling implemented as a differentiable gather with the index map of the verifier, and

$$H_\beta(\rho) = \frac{\tanh(\beta\eta) + \tanh(\beta(\rho - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \quad \eta = 0.5, \quad (2)$$

the smoothed Heaviside projection [20]. Thirty-two L-BFGS steps [37] (step size 0.5, strong-Wolfe line search) are split evenly across $\beta \in \{1, 4, 16, 64\}$, and L-BFGS is restarted at each level because curvature information from a softer projection does not carry over. The continuous baseline applies the loss to $D(z, c)$ with bilinear resampling for 30 steps. Guarded refinement keeps the refined design only if its verified joint error is smaller than that of the start.

Manufacturability-aware refinement (Section 3.5) adds $\lambda [P_c(x) + P_t(x)]$ to $\mathcal{L}_\beta$, evaluated on the projected image $x = H_\beta \circ D(z, c)$ at the 64×64 resolution on which manufacturability is checked. P_c is the mean over all periodic 2×2 windows of $x_{11}x_{22}(1 - x_{12})(1 - x_{21}) + x_{12}x_{21}(1 - x_{11})(1 - x_{22})$, which counts corner-only contacts on a binary image; P_t is the mean of $x - \text{open}(x)$, where the morphological opening uses the cross-shaped element of the feature-size check, so it is non-zero exactly on struts thinner than three pixels (and, weakly, on convex corners). Disconnected islands are not differentiable and are left to the verifier. The weight λ was tuned on 20 targets of set B and fixed before sets A and C were run.

2.6 Topology optimization baseline and hybrid

For each target we minimized $\frac{1}{2} \|\nu(\hat{\mathbf{x}}) - \nu^*\|_2^2$ over element densities $\mathbf{x}$, with $\hat{\mathbf{x}} = H_\beta(\tilde{\mathbf{x}})$ and $\tilde{\mathbf{x}}$ the density-filtered field (radius 1.5 elements), subject to $\tilde{\mathbf{x}} \leq 0.55$ and $Q_{11}, Q_{22} \geq 0.1 \cdot 0.55 E_0$, using the method of moving asymptotes [28] with penalization $p = 3$. Sensitivities follow from the same analytic expressions used for refinement. Continuation runs $\beta \in \{1, 2, 4, 8, 16, 32, 64\}$ with at most 43 evaluations per level and a restart of the optimizer at each level. The four seed topologies of the training data serve as starting points, rescaled to the volume bound. The final $\hat{\mathbf{x}}$ is thresholded at 0.5 and verified with an independent FE solve, exactly like a generated design, and the best of the four starts is selected by verified joint error. Process parameters are the medians of the training data. A budget-matched variant allows at most 105 evaluations per start. The hybrid uses the generated binary design as the starting point and $\beta \in \{8, 16, 32, 64\}$ with at most 15 evaluations per level.

2.7 Retrieval and out-of-distribution protocols

Retrieval returns the training cells with the smallest Euclidean distance to the target in (ν_{12}, ν_{21}) label space; each retrieved image is verified by the same chain as a generated one, either the nearest cell alone or the best of the 30 nearest by verified error. For targets with $\nu_{12}^* > 0$ (sign-flip test) we used 28 targets in three groups: symmetric $\nu_{12}^* = \nu_{21}^* \in \{0.05, \dots, 0.60\}$ (12 targets) and $\nu_{12}^* \in \{0.05, \dots, 0.80\}$ with $\nu_{21}^* = 0.1$ or 0.3 (8 targets each). The magnitude sweep used six anisotropic targets with $\nu_{21}^* = -0.05$ and ten independent repeats of best-of-30 per target. Admissibility was checked against $\nu_{12}\nu_{21} < 1$ [38].

2.8 Statistics and pre-specification

The in-distribution target sets A, B and C consist of 100 test conditions each, drawn with seeds 123, 456 and 789 (4 conditions shared between A and B). Relative MAE reductions and 95% CIs are computed by paired percentile bootstrap over conditions with 10 000 resamples [39]. An improvement is claimed only if the CI excludes zero. Success criteria were written into the project plan before each experiment was run (pre-specified; the plan is version-controlled in the code repository). They were: at least 20% MAE

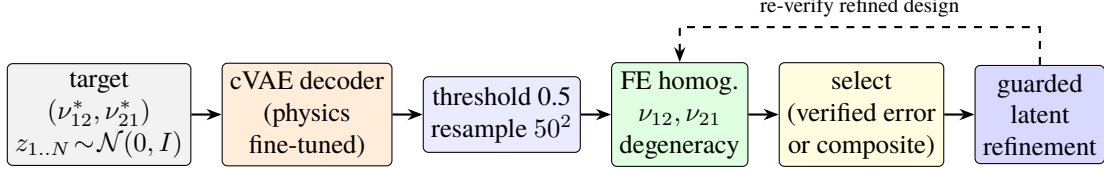


Figure 1: Inverse-design pipeline. Every candidate is verified by FE homogenization after the same post-processing that a manufactured design would undergo. Refinement is detailed in Fig. 2.

reduction for single samples; no significant MAE increase after best-of-30; a median relative error of at most 8% at $\nu_{12}^* = -2.0$; a mean Spearman correlation of at least 0.7 for the composite score; for KAN, a CI excluding zero for both seeds; for WIRE, a hit rate of at least 30% with FE $R^2 > 0$; for the hybrid, joint-error non-inferiority within 10% against converged topology optimization, at least 70% manufacturable, at most 299 FE solves per target and at most 1% degenerate designs; and for the stratified comparison on set C, CIs excluding zero in favour of the generative pipeline for both the joint and the ν_{12} error at anisotropy ratio $r < 10$; and for manufacturability-aware refinement, on set A, at least 60% manufacturable (4-connectivity), $R^2(\nu_{12}) \geq 0.99$, $R^2(\nu_{21}) \geq 0.80$, at most 13% of designs with corner-only contacts and at most 10% more FE solves, with the first two replicated on set C. Sets A and B were analysed before the stratification was conceived, so their stratified results are post hoc. The refinement method was developed after continuous refinement had failed on set A; sets B and C serve as its confirmation. L-BFGS on GPU is not bit-wise deterministic, and repeated runs differed by about 0.006 in R^2 .

3 Results

3.1 Pipeline and evaluation protocol

Figure 1 shows the pipeline. A cVAE maps a target pair (ν_{12}^*, ν_{21}^*) and latent vectors $z \sim \mathcal{N}(0, I)$ to 64×64 density images. Each is binarized, resampled to the FE grid and homogenized. The best candidate is refined in latent space, and the refined design is kept only if its verified error is lower (“guarded” refinement). All numbers below are FE-verified on the binarized design.

3.2 Verified accuracy requires optimizing the verified design

Continuous refinement exploits grey densities. Refining the continuous decoder output (30 L-BFGS steps, gradients through the decoder and the exact homogenization sensitivities) helps single samples of the production model. On set A, $R^2(\nu_{12})$ rises from 0.874 to 0.976 and the mean absolute error (MAE) of ν_{12} falls by 57.7% (95% CI 49.4–64.6%; Table 1, Fig. 3b). The same refinement, however, *worsens* the best-of-30 winner: MAE(ν_{12}) rises from 0.0142 to 0.0247, a change of −73.8% (CI −119 to −39%), and R^2 drops from 0.988 to 0.962. The objective trace explains why (Fig. 4). At the end of continuous refinement the median objective is 2.5×10^{-10} , but the median verified squared error of the binarized design is 4.4×10^{-4} . The optimizer matches the target on a field whose intermediate densities do not survive thresholding (Fig. 5). This is the failure mode of surrogate exploitation, arising inside an exact physics model.

Binarization-aware refinement. Placing differentiable counterparts of the verification operators inside the objective (Fig. 2) shrinks the median ratio of verified error to objective from about 10^6 to about 2×10^2 (Fig. 4). All pre-specified in-distribution criteria are met (Table 1).

- **Single samples.** $R^2(\nu_{12})$ rises from 0.874 to 0.999 and MAE(ν_{12}) falls by 92.7% (CI 90.2–94.6%). The mean relative error of ν_{12} drops from 14.4% to 2.3%, and $R^2(\nu_{21})$ rises from 0.477 to 0.875.
- **After best-of-30.** MAE(ν_{12}) falls by 68.3% (CI 59.3–75.8%), $R^2(\nu_{12})$ rises from 0.988 to 0.998, and MAE(ν_{21}) falls from 0.030 to 0.010 ($R^2(\nu_{21})$ 0.812 to 0.845). Guarded refinement accepts 94%

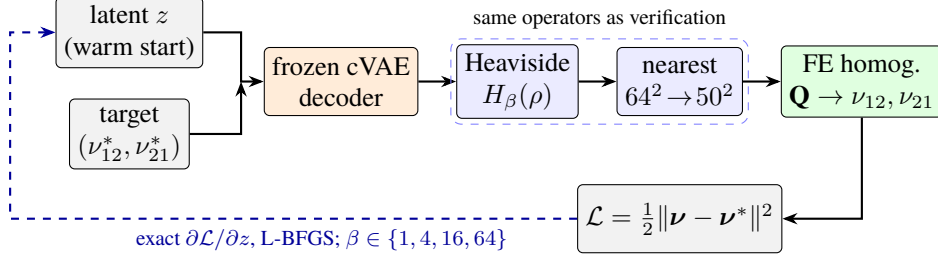


Figure 2: Binarization-aware latent refinement. The frozen decoder output passes through differentiable counterparts of the verification operators before FE homogenization, so the quantity optimized is the quantity verified. The continuous variant omits the dashed block and uses bilinear resampling. Verification replaces H_β by a hard threshold at 0.5.

Table 1: FE-verified accuracy on set A (100 held-out targets, production cVAE). ΔMAE is the relative reduction of $\text{MAE}(\nu_{12})$ versus the unrefined design of the same mode, with a paired bootstrap 95% CI. Negative values mean that refinement made results worse. Joint: mean of $|\Delta\nu_{12}| + |\Delta\nu_{21}|$. FE: solves per target, including selection.

Mode	Refinement	$R^2(\nu_{12})$	$\text{MAE}(\nu_{12})$	$R^2(\nu_{21})$	Joint	ΔMAE [95% CI]	FE
Single	none	0.874	0.0517	0.477	0.127	—	1
	continuous	0.976	0.0219	0.643	0.053	+57.7% [49.4, 64.6]	51
	binarization-aware	0.999	0.0038	0.875	0.012	+92.7% [90.2, 94.6]	78
Best-of-30	none	0.988	0.0142	0.812	0.044	—	30
	continuous	0.962	0.0247	0.671	0.052	−73.8% [−119, −39]	79
	binarization-aware	0.998	0.0045	0.845	0.014	+68.3% [59.3, 75.8]	107
	guarded	0.998	0.0046	0.846	0.014	+67.4% [57.9, 75.3]	107

of refinements and gives the same accuracy.

Refinement costs 77 FE solves per target on average, i.e. about 7 s of CPU time on the 50×50 grid. Guarding alone does not rescue continuous refinement after best-of-30 (MAE reduction 10.5%, CI −3.0 to 22.4%).

Replication on fresh target sets. Because the method was developed after continuous refinement had failed on set A, we repeated it on sets B and C, drawn with different seeds. For single samples on set B, $R^2(\nu_{12})$ rises from 0.831 to 0.989 and $\text{MAE}(\nu_{12})$ falls by 87.8% (CI 82.3–92.3%). After best-of-30, $\text{MAE}(\nu_{12})$ falls by 59.3% (CI 46.8–70.9%) on set B and by 63.6% (CI 46.3–75.2%) on set C.

3.3 Comparison with topology optimization from scratch

The relevant competitor of a generative pipeline is not a lookup table but topology optimization run directly for the target. We solved the classical inverse homogenization problem for every target from four starting topologies (Section 2.6), verified each result exactly like a generated design, and kept the best of the four by verified error. This baseline uses on average 893 FE solves per target (median 102–140 s of CPU time on one core), against 107 for best-of-30 sampling followed by guarded binarization-aware refinement.

Topology optimization shows the same gap. At the end of continuation the median error on the projected density field was 2×10^{-7} , whereas the median verified error after thresholding was 3.6×10^{-2} per run, and for 57% of the 300 targets the starting topology with the lowest projected error was not the one with the lowest verified error. Selection by verified error is therefore necessary for both methods.

Accuracy, cost and failure mode. Table 2 compares the two methods on sets A, B and C. On all three, the generative pipeline matches converged topology optimization in ν_{12} ($R^2 = 0.995$ – 0.998 versus 0.996 – 0.998) with roughly eightfold fewer FE solves. Over all targets its joint error is not consistently lower,

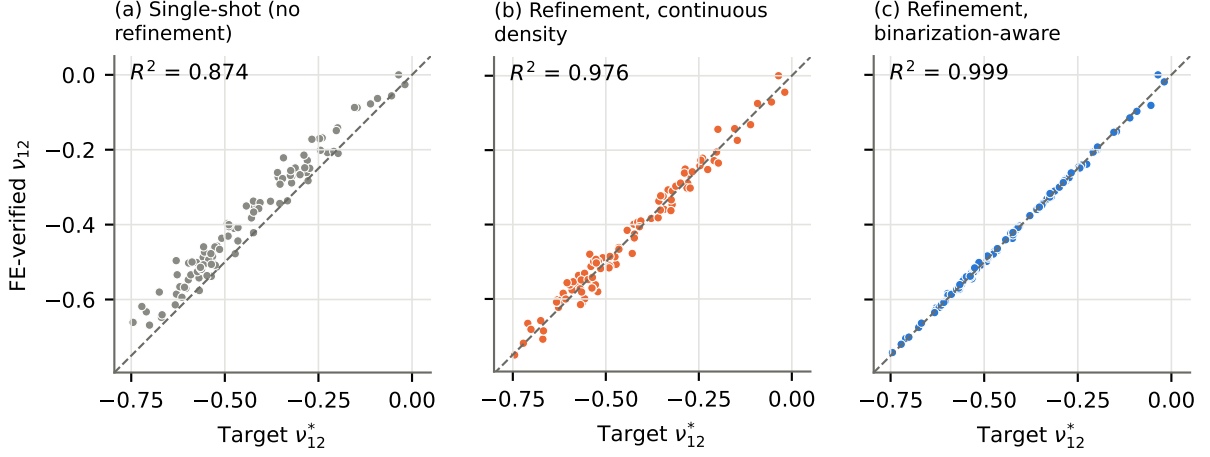


Figure 3: FE-verified versus target ν_{12} on set A ($n = 100$, single samples of the production cVAE): (a) unrefined; (b) after continuous refinement; (c) after binarization-aware refinement. All panels start from the same latent code per condition. Dashed line: identity.

because of one failure mode. Every degenerate cell produced by the cVAE (one in set A, two in set B, none in C) belongs to a target with an anisotropy ratio $r = \max(\nu_{21}^*/\nu_{12}^*, \nu_{12}^*/\nu_{21}^*) \geq 10$, which corresponds to a stiffness ratio $E_2/E_1 \geq 10$ for an orthotropic cell. In these cells one column of pixels is empty, so the cell carries no load in one direction and its Poisson ratio is meaningless (Fig. 6). For these targets all 30 sampled candidates were disconnected, so the failure cannot be fixed by selection; only 2.8% of the training cells have $r > 10$. Without the three degenerate targets, the joint error of the generative pipeline is 0.0075, 0.0091 and 0.0082 against 0.0127, 0.0140 and 0.0126 for topology optimization, which produced no degenerate cell. The low $R^2(\nu_{21})$ of the generative pipeline on sets A and B (0.846, 0.745) is caused by the same targets.

Stratified, pre-specified confirmation. Having observed this on sets A and B, we pre-specified a stratified test on set C before running it: for targets with $r < 10$, the generative pipeline should have a lower joint error and a lower ν_{12} error than topology optimization, with paired 95% CIs excluding zero. This test was run with the earlier pipeline that still enforced periodic edges (Appendix A). The joint-error criterion was met (27% lower, CI 8–42%); the ν_{12} criterion was not (19% lower, CI –10 to 40%). With the final pipeline, both criteria are met on all three sets (Table 2); because this version was not pre-specified, we report it as a sensitivity analysis and claim parity in ν_{12} and an advantage in joint error for moderate anisotropy, not superiority in ν_{12} . Of the five targets with $r \geq 10$, the generative pipeline was better on one (set C) and much worse on four.

Mesh sensitivity and stiffness. All accuracies above are relative to the 50×50 FE model used for data generation, refinement and verification. Re-solving 30 test geometries with each pixel subdivided into 4×4 elements changes ν_{12} by a median of 0.015 (3.8%) and at most 0.042, towards more negative values in 27 of them. The differences in ν_{12} between the two methods ($\sim 10^{-3}$) are therefore an order of magnitude smaller than the discretization error of the model. On the 200×200 mesh the ranking of the two methods on set A is unchanged ($R^2(\nu_{12})$ 0.990 versus 0.985, MAE 0.014 versus 0.016). The auxetic designs of both methods are not near-mechanisms: the median in-plane modulus is $E_1 \approx 0.07\text{--}0.08 E_0$ for both (Fig. 7).

A hybrid does not inherit the strengths of both. Because topology optimization was more robust and more manufacturable while the generative pipeline was cheaper, we pre-specified a hybrid in which the generated design initializes topology optimization (continuation $\beta = 8 \rightarrow 64$, at most 61 additional FE solves). It met the cost and non-degeneracy criteria (166 FE solves per target; 1% degenerate) but failed the joint-error non-inferiority criterion against converged topology optimization (CI lower bound –96%,

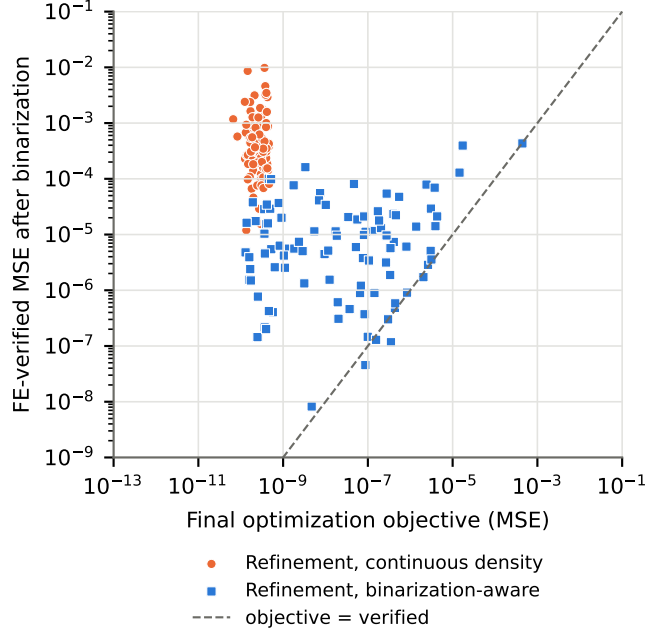


Figure 4: Objective-verification gap. Each point is one condition of set A. Horizontal axis: final refinement objective; vertical axis: squared Poisson-ratio error of the binarized design measured by an independent FE solve. Continuous refinement (circles) reaches objectives near 10^{-10} while verified errors remain at 10^{-4} – 10^{-2} . Binarization-aware refinement (squares) moves points towards the identity. Values below 10^{-14} are clipped for display.

driven by the one degenerate target) and the manufacturability criterion (0.33 against a threshold of 0.70). Starting from a sharp projection preserves the topology, including the thin features, of the generated design. This experiment was run with the earlier pipeline with periodic-edge enforcement.

Cost. The saving of about 790 FE solves per target pays off the $2.7\text{--}3.0 \times 10^6$ FE solves spent on the training data only after about 3 500–3 800 targets (11 500–12 700 against the budget-matched variant of topology optimization). The cost argument therefore holds when many targets must be served, not for a single design.

3.4 Retrieval from the training library

A lookup in the training library is the cheapest alternative. Verified by the same chain, the single nearest cell reaches $R^2(\nu_{12}) = 0.941$ on set A and the best of the 30 nearest cells $R^2 = 0.977$ with a joint error of 0.035, slightly better than best-of-30 generation without refinement (0.988, joint error 0.044). Binarization-aware refinement lifts the generative pipeline to $R^2 = 0.998$ and a joint error of 0.014. The verified properties of a training cell also differ from its label: resampling from 64^2 to 50^2 , penalization and binarization together shift ν_{12} by an MAE of 0.032 on 150 test cells, because the labels were computed on the grey SIMP fields. Labels are thus another quantity that differs from what is verified.

3.5 Manufacturability and multi-objective selection

Accurate designs are not automatically buildable. With best-of-30 selection by verified error and guarded refinement, 32–35% of the final generated designs are manufacturable, against 74–83% for topology optimization (Table 2). Refinement does not reduce manufacturability: over four runs (single samples and best-of-30 on set A, best-of-30 on sets B and C), the manufacturable fraction rose by 4–7 percentage points, with 55 designs becoming manufacturable and 33 ceasing to be (sign test $p = 0.025$; two of the runs share set A).

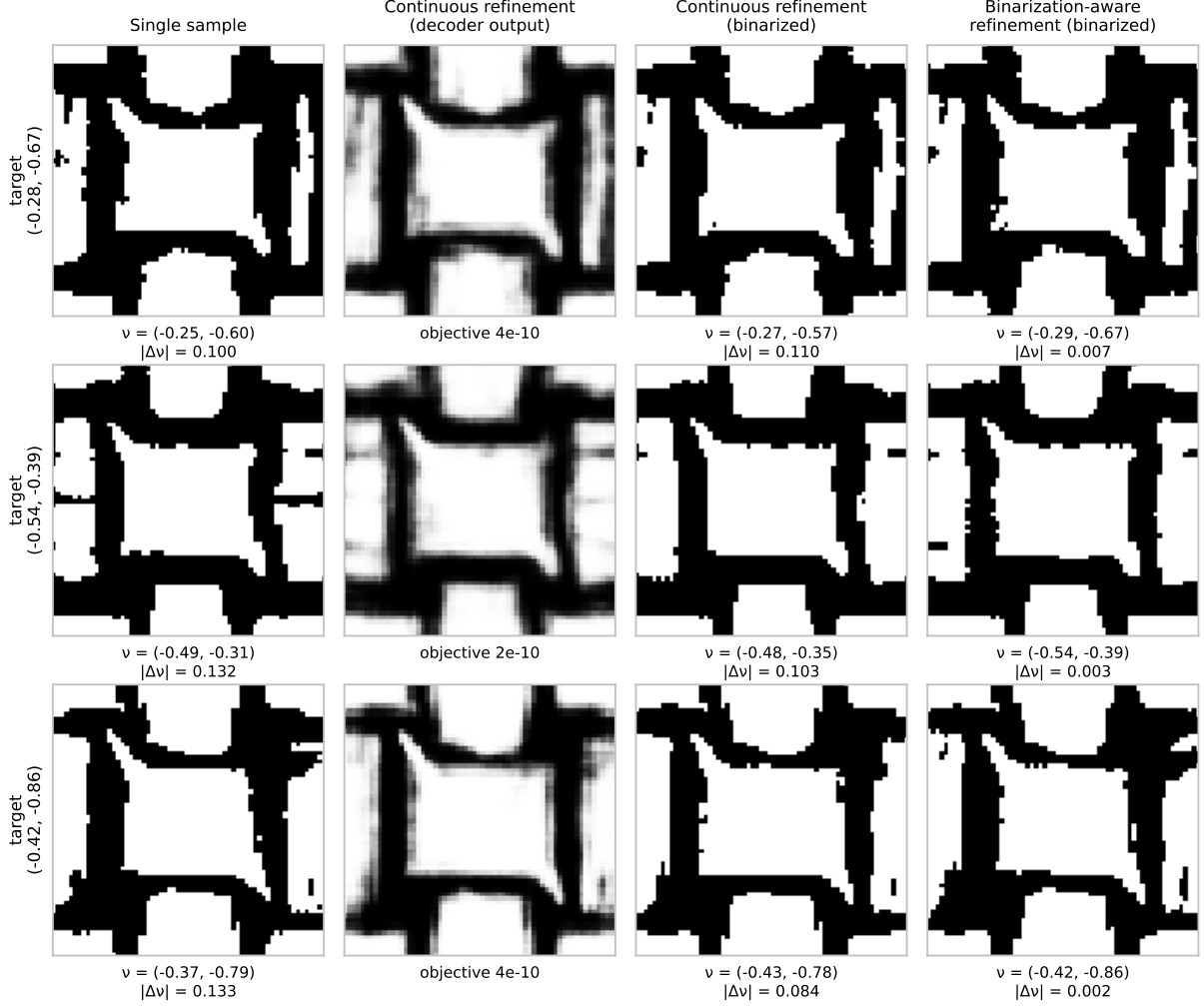


Figure 5: The same starting latent code for three conditions of set A: single sample, continuous refinement (decoder output and its binarization), and binarization-aware refinement (binarized). Continuous refinement reaches its target on a grey field whose binarization has different properties.

Manufacturability-aware refinement. Following the principle of optimizing what is verified, we added the corner-contact and thin-feature penalties of Section 2.4 to the refinement objective ($\lambda = 0.1$) and pre-specified the criteria listed in Section 2.8. All were met. With 4-connectivity, the manufacturable fraction of the final designs rose from 0.31 to 0.72 on set A (paired difference +0.41, CI 0.31–0.51) and from 0.31 to 0.76 on set C (+0.45, CI 0.34–0.56), against 0.72 and 0.80 for topology optimization (8-connectivity: 0.72 and 0.77 against 0.77 and 0.83). Designs with corner-only contacts fell from 16% to 8% (set A) and from 21% to 5% (set C); FE solves per target rose from 107 to 112, and $R^2(\nu_{12})$ remained 0.996–0.997. The gain is not free. The MAE in ν_{12} increased by 37% (CI 5–78%) on set A and by 40% (CI 0–98%) on set C, and the joint-error advantage over topology optimization for targets with $r < 10$ shrank from 41% to 15% (CI –10 to 34%) on set A and from 35% to 14% (CI –7 to 32%) on set C, i.e. it is no longer significant. A tuning run on 20 targets had suggested that the penalties also improve accuracy; this did not replicate on 100. The weight λ therefore selects an operating point rather than a better method (Fig. 8): $\lambda = 0$ for the accuracy advantage reported above, $\lambda = 0.1$ for manufacturability close to that of topology optimization at statistically indistinguishable joint error and about eight times fewer FE solves. On all 100 targets of set B, the manufacturable fraction rises steeply up to $\lambda = 0.1$ (0.33, 0.61, 0.79 for $\lambda = 0, 0.03, 0.1$) and saturates beyond (0.81 and 0.83 for 0.3 and 1), while the MAE in ν_{12} grows from 0.0059 to 0.0066–0.0068 for $0.03 \leq \lambda \leq 0.3$ (increase 11–16%, CIs including zero on this set) and to 0.0077 at $\lambda = 1$ (30%, CI 6–64%); $\lambda = 0.1$ sits at the knee.

Table 2: Generative pipeline (best-of-30 + guarded binarization-aware refinement) versus topology optimization from scratch (best of four starts, converged) on three disjoint sets of 100 held-out targets. Both methods select by verified joint error. Joint: mean $|\Delta\nu_{12}| + |\Delta\nu_{21}|$. FE: solves per target. Deg.: degenerate cells ($\min E_i < 10^{-3} E_0$). Manuf.: fraction manufacturable. Bottom rows: relative error reduction of the generative pipeline versus topology optimization for targets with anisotropy ratio $r < 10$ (paired bootstrap 95% CI; positive favours the generative pipeline).

Set	Method	$R^2(\nu_{12})$	$R^2(\nu_{21})$	Joint	FE	Deg.	Manuf.
A	generative	0.998	0.846	0.0139	107	1	0.32
	topology opt.	0.997	0.997	0.0128	897	0	0.77
B	generative	0.995	0.745	0.0229	107	2	0.35
	topology opt.	0.996	0.997	0.0139	889	0	0.74
C	generative	0.998	0.998	0.0082	107	0	0.32
	topology opt.	0.998	0.996	0.0126	893	0	0.83
$r < 10$, joint error: A +41% [26, 54] B +40% [23, 53] C +35% [19, 48]							
$r < 10$, ν_{12} error: A +39% [19, 54] B +39% [19, 54] C +37% [14, 54]							

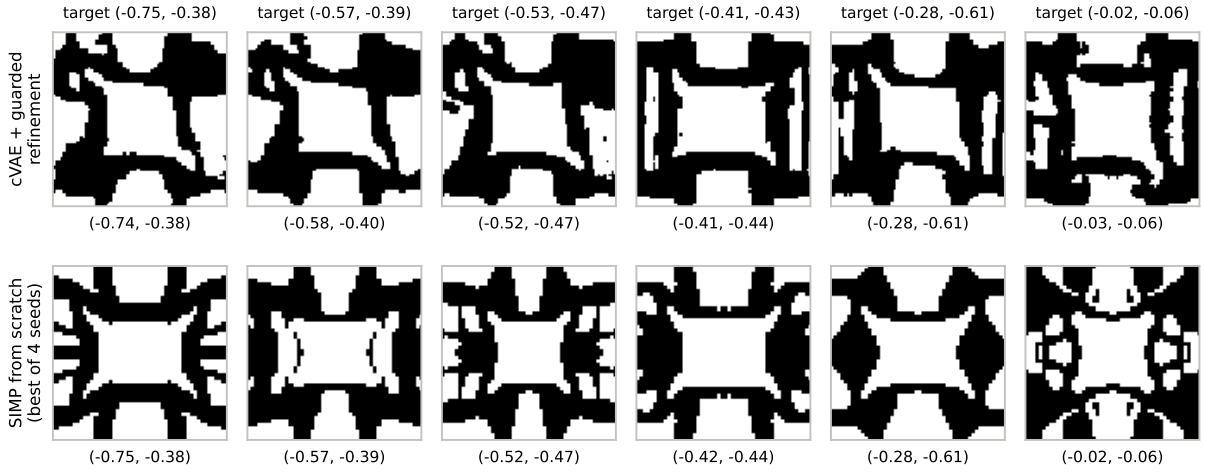


Figure 6: Final designs of the generative pipeline and of topology optimization from scratch for the same targets of set A, with target and verified (ν_{12}, ν_{21}) .

To trade accuracy against manufacturability during selection, the composite score of Section 2.4 can replace the verified error. On 300 test targets ($N = 30$), composite selection keeps $R^2(\nu_{12}) = 0.988$ (0.997 with accuracy-only selection) while every first sample is auxetic; 23.8% of all generated samples are manufacturable (Table 3). This R^2 differs from that of Table 1 because the targets and the latent draws differ. To test whether the scalarization respects the multi-objective structure, we computed, for 24 targets with 30 candidates each, the Spearman correlation between the composite score and the non-dominated sorting rank over the three raw objectives (Fig. 9). All 24 correlations were positive (0.45–0.86, median 0.746, mean 0.718). The pre-specified criterion (mean ≥ 0.7) was set for, and failed with, the earlier pipeline (mean 0.693); we therefore read the composite score as agreeing moderately, not strongly, with the Pareto structure. That the selected candidate lies on the first Pareto front is guaranteed for positive weights and is not evidence.

Figure 10 shows that the generated cells tile without edge matching, and Fig. 11 shows the deformation mechanism of a generated auxetic cell and of a generated cell with $\nu_{12} > 0$.

3.6 Out-of-distribution targets

Admissible targets. For a two-dimensional orthotropic material, positive definiteness of the compliance requires $\nu_{12}\nu_{21} < 1$ [38]. Symmetric targets $\nu_{12}^* = \nu_{21}^* \leq -1$ are therefore infeasible. In the training

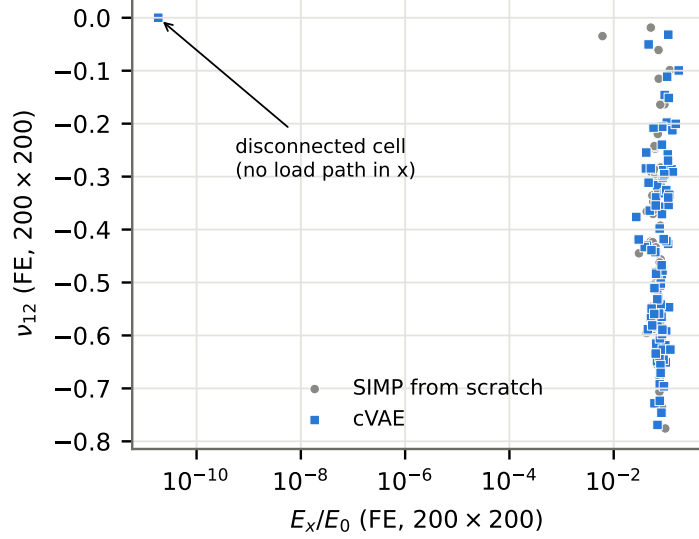


Figure 7: Verified ν_{12} versus E_1/E_0 of the final designs on set A, both on the 200×200 mesh.

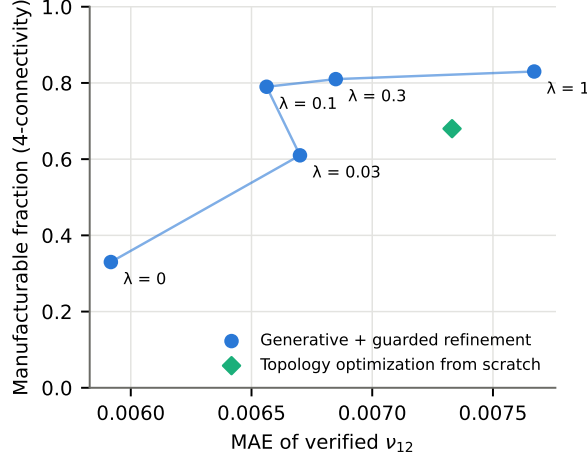


Figure 8: Accuracy–manufacturability trade-off of manufacturability-aware refinement on set B (100 targets, best-of-30, guarded), for penalty weights λ from 0 (refinement of Section 3.2) to 1, and topology optimization from scratch on the same targets. Manufacturable: 4-connectivity and minimum feature size.

data the most negative $\nu_{12} = -1.95$ is paired with $\nu_{21} = -0.04$, and the largest product $\nu_{12}\nu_{21}$ is 0.43. An earlier “extreme auxetic” test in our project used targets such as $(-2.1, -2.1)$, all of which violate the bound; we exclude such targets. Out-of-range magnitude targets below are anisotropic, with $\nu_{21}^* = -0.05$.

Sign: the generator reaches unseen signs, but topology optimization is more accurate. Every training cell is auxetic ($-1.95 \leq \nu_{12} \leq -0.002$), so retrieval cannot return $\nu_{12} > 0$; for the 28 sign-flip targets its error is at least 0.33–0.37 on average. Best-of-30 generation followed by guarded binarization-aware refinement produced the correct sign for most targets and reduced the MAE of ν_{12} by 66–73% relative to best-of-30 alone (Table 4, Fig. 12). Topology optimization from scratch, however, was markedly more accurate in all three groups, also when only targets with non-degenerate designs from both methods are compared. Both methods produced degenerate designs (three each). The three degenerate cells from topology optimization are empty: with all elements at $E_{\min}$, the cell has the Poisson ratio of the base material, 0.3, and an unguarded verifier scores it as a near-perfect answer to targets between 0.25 and 0.30. Only 8–12% of the generated cells for these targets were manufacturable, against 75% for topology optimization.

Table 3: Selection on 300 test targets (best-of-30, no refinement). Hit rate: fraction of targets whose first sample is auxetic ($\nu_{12} < 0$). Manufacturable: share of all generated samples.

Selection	$R^2(\nu_{12})$	Hit rate, first sample	Manufacturable samples
Verified accuracy only	0.997	1.00	0.238
Composite score	0.988	1.00	0.238
Composite score vs. Pareto rank (24 targets $\times$ 30 candidates): Spearman mean 0.718, median 0.746, range 0.45–0.86; 24/24 positive.			

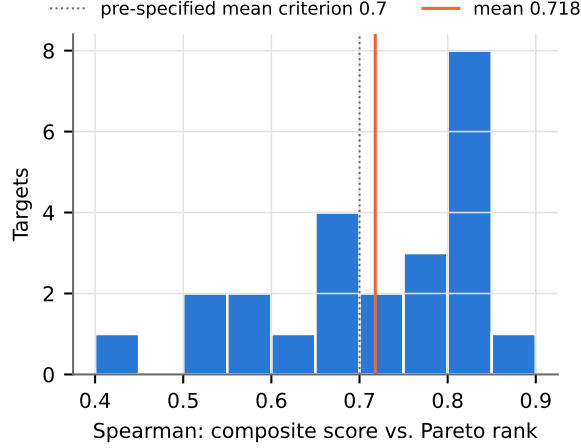


Figure 9: Distribution over 24 targets of the Spearman correlation between the composite score and the Pareto rank of 30 candidates (objectives: verified $|\Delta\nu_{12}|$, manufacturability score, aesthetic score). Dotted: pre-specified threshold for the mean.

Magnitude: refinement extends into sparse data but not beyond. We swept $\nu_{12}^* \in \{-1.0, -1.5, -1.75, -2.0, -2.25, -2.5\}$ ($\nu_{21}^* = -0.05$) with ten independent latent draws per target and best-of-30 selection (Table 5, Fig. 13). The training data thin out well before their minimum: 88 training samples have $\nu_{12} < -1.5$, 14 have $\nu_{12} < -1.75$ and two have $\nu_{12} < -1.9$, all of them 90° rotations of cells with extreme ν_{21} . Binarization-aware refinement reaches a median relative error of 0.3% at -1.75 and 1.2% at -2.0 , which meets the pre-specified 8% criterion at -2.0 . The distribution there is bimodal, however: six of the ten repeats end below 1.7%, four between 17% and 31%. From -2.25 onwards every variant saturates, and unguarded refinement is worse than no refinement. Refinement is a local search in the latent space of the decoder and cannot create magnitudes the decoder never learned to represent.

3.7 Architecture ablations

KAN versus linear heads. We replaced the three linear layers of the cVAE (latent mean, log-variance and decoder bottleneck) with Kolmogorov–Arnold layers [40]. Both heads were trained with identical recipes, hyperparameters and seeds (123 and 7). KAN was declared the winner only if the paired CI excluded zero for both seeds. It did not win in any mode (Table 6). In the production mode (best-of-30 without refinement), the linear head was significantly better for both seeds. With refinement, both heads reached $R^2(\nu_{12}) = 0.997$ – 0.999 . The KAN head has $3.2\times$ more parameters (4.79 M versus 1.49 M) and needs about $1.6\times$ the time per epoch, so we retain the linear head. This ablation was evaluated with the earlier pipeline with periodic-edge enforcement, which does not change accuracy on set A (Appendix A).

Implicit wavelet decoder. A resolution-agnostic WIRE decoder [41] was given one final attempt. It combined Heaviside projection inside the physics loss, no forced symmetry and the two-stage recipe. The pre-specified criteria were a single-sample hit rate of at least 30% and a positive FE R^2 on 24 held-out targets. It reached a hit rate of 12.5% and an FE R^2 of -7.19 after best-of-30, against a hit rate of 100%

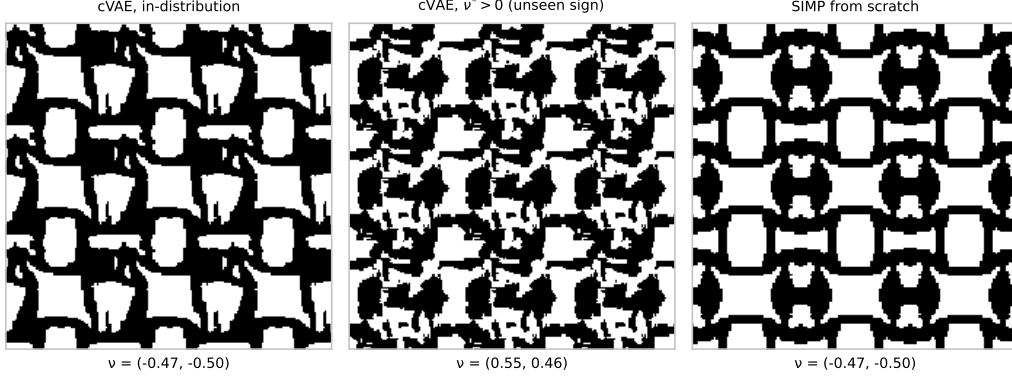


Figure 10: 3×3 tilings of final designs: generative pipeline in distribution, generative pipeline for an unseen sign ($\nu_{12}^* > 0$), and topology optimization from scratch. Periodic boundary conditions on the element mesh make every pixel cell tile by translation.

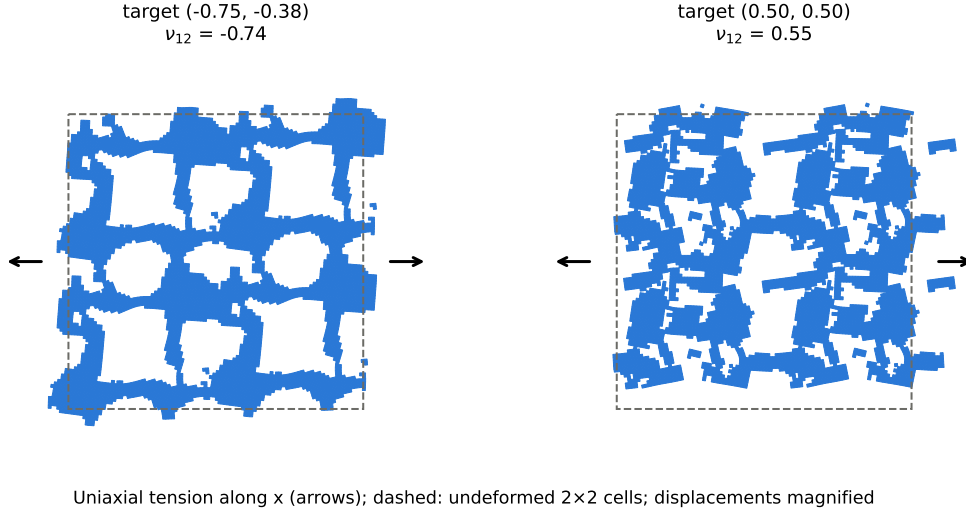


Figure 11: Deformation under uniaxial tension along x (FE, 50×50 , displacements magnified) of a generated auxetic cell (target $(\nu_{12}^*, \nu_{21}^*) = (-0.75, -0.38)$, verified $\nu_{12} = -0.744$) and of a generated cell for $\nu_{12}^* = \nu_{21}^* = 0.5$ (verified 0.547). Dashed: the undeformed 2×2 cells. The two solid regions of the second cell touch only at pixel corners (one-node hinges); it is not manufacturable under 4-connectivity although $E_1/E_0 = 0.016$ exceeds the degeneracy threshold.

for the convolutional decoder, and only 4.6% of its samples were manufacturable. We therefore closed this direction.

4 Discussion

The generative pipeline as an inverse-design solver. Measured against converged topology optimization rather than against a lookup table, the advantage of the generative pipeline is narrower than is commonly implied, in line with the critical review of Woldseth et al. [17], but it is real. It reaches the same verified accuracy with about eight times fewer FE solves per target, and a lower joint error for targets of moderate anisotropy, but the saving is amortized only after thousands of targets because the training data themselves were produced by topology optimization. Without refinement, retrieval from the same library is about as accurate. The pipeline is therefore attractive when many targets must be served, for example in interactive design or in the inner loop of a multiscale optimization, and not when a single cell is needed. Its failures are predictable: strongly anisotropic targets that are sparse in the training data. Reporting such

Table 4: Sign-flip targets ($\nu_{12}^* > 0$; training data contain only $\nu_{12} < 0$). MAE of FE-verified ν_{12} : retrieval (lower bound, since every training label is ≤ -0.002), generative pipeline before and after guarded refinement, and topology optimization (TO). In brackets: MAE over targets for which neither method returned a degenerate design (n in the last column). Deg.: degenerate designs, generative / TO.

Group	Retrieval	Best-of-30	+ refinement	TO	Deg.	n
$\nu_{12}^* = \nu_{21}^* (12)$	≥ 0.33	0.204	0.056 [0.030]	0.012 [0.009]	1 / 2	9
$\nu_{21}^* = 0.1 (8)$	≥ 0.37	0.272	0.090 [0.054]	0.031 [0.032]	1 / 0	7
$\nu_{21}^* = 0.3 (8)$	≥ 0.37	0.222	0.076 [0.033]	0.005 [0.006]	1 / 1	6
FE per target	–	30	105–107	761–826		

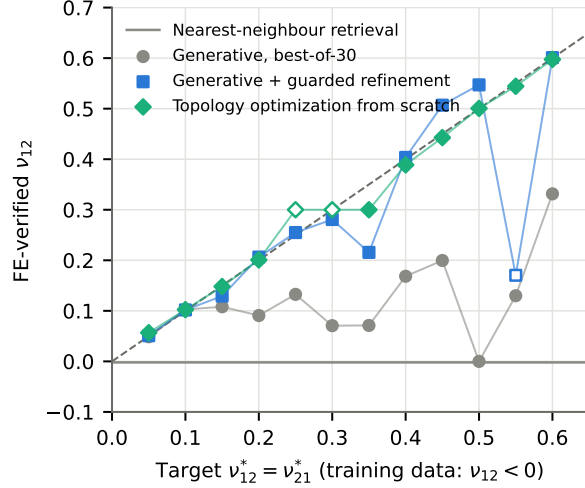


Figure 12: Symmetric sign-flip targets: FE-verified ν_{12} of the generative pipeline (best-of-30 and after guarded refinement) and of topology optimization from scratch. Retrieval is bounded by the training library ($\nu_{12} \leq -0.002$, grey line). Open markers: degenerate designs. Dashed: identity.

failures by stratum, rather than averaging them into a single score, made them visible in our study.

Choosing a method. Table 7 condenses the comparison into a rule for the designer. Each row is supported by a result in Section 3.3 or later; none rests on a single run.

Optimizing what is verified. The accuracy of the pipeline rests on one finding: any gradient-based search over density fields, whether in the latent space of a generative model or over element densities in topology optimization, will exploit the difference between the field it differentiates and the binarized design that is finally analysed. The optimizer reaches objectives of 10^{-10} on fields that the verifier sees very differently (Fig. 4), and the same happens in topology optimization with Heaviside continuation to $\beta = 64$. The gap is invisible in the loss curve. Three remedies follow, and all are cheap: include the verification operators in the objective, select among candidates by verified rather than optimized error, and treat degenerate cells as failures in the verifier itself. Neither of the first two ingredients is new in density-based topology optimization [19, 20]; what this work shows is that omitting them changes the conclusions of a comparison between methods, and that the generative and the classical route need them equally. The third is easy to overlook: an empty cell has the Poisson ratio of its base material, and in our sign-flip test such cells were the best answers that topology optimization returned for three targets.

Manufacturability and out-of-distribution targets. Topology optimization from scratch produced connected cells with adequate feature size more than twice as often as the generative pipeline when only the Poisson ratios were optimized. Initializing topology optimization with the generated design did not recover

Table 5: Anisotropic magnitude sweep ($\nu_{21}^* = -0.05$; best-of-30; 10 repeats per target). Entries: median relative error of FE-verified ν_{12} . The training range of ν_{12} ends at -1.95 .

ν_{12}^*	-1.0	-1.5	-1.75	-2.0	-2.25	-2.5
Best-of-30, no refinement	0.5%	0.8%	3.6%	10.1%	15.6%	21.7%
+ continuous refinement	1.3%	1.2%	3.5%	31.0%	41.6%	45.5%
+ binarization-aware refinement	0.4%	0.2%	0.3%	1.2%	24.7%	28.8%
+ binarization-aware, guarded	0.1%	0.2%	0.3%	1.2%	14.2%	20.0%

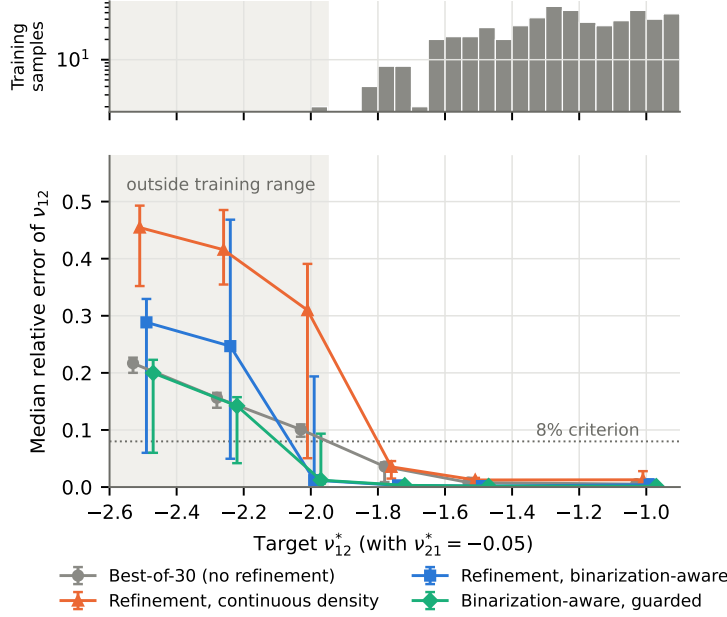


Figure 13: Top: number of training samples per bin of ν_{12} (log scale). Bottom: median relative error of FE-verified ν_{12} versus target ($\nu_{21}^* = -0.05$). Error bars: interquartile range over 10 latent draws per target; markers offset horizontally for legibility. Shaded: outside the training range. Dotted: pre-specified 8% criterion at $\nu_{12}^* = -2.0$.

that fraction, because a sharp initial projection preserves the generated topology; penalizing the verified manufacturability defects during latent refinement did, at the price of the accuracy advantage. Accuracy and manufacturability compete for the same few latent degrees of freedom, and the weight between them is a design decision that the verifier, not the loss curve, has to inform. For non-auxetic targets the cVAE reaches the correct sign where retrieval from the library cannot, but topology optimization reaches the target far more accurately. Along magnitude, refinement pushes into regions with only a handful of training samples, but all generative variants stop where the data end. OOD claims for generative models should therefore name the direction in property space, check that the target is physically admissible, and compare against topology optimization rather than retrieval. Magnitude extrapolation will likely require new training data near the boundary.

Limitations. (i) Accuracy is measured against a 50×50 linear-elastic, small-strain FE model with $\nu_0 = 0.3$; its discretization error (median 0.015 in ν_{12}) exceeds the reported mean errors. Large strain, other base materials and three-dimensional cells are not addressed, and no design was tested experimentally. (ii) The generative pipeline fails for strongly anisotropic targets ($r \geq 10$), which are sparse in the training data; topology optimization does not. (iii) Without the manufacturability penalty, topology optimization produces manufacturable cells more than twice as often (0.74–0.83 versus 0.32–0.35); with it, the generative pipeline comes close (0.72–0.77) but loses its joint-error advantage. Manufacturability is checked geometrically, not by a printing process model. (iv) For targets with an unseen sign, topology optimization is markedly more accurate; the advantage of the generative model in that regime holds only

Table 6: Controlled KAN versus linear ablation on set A (two seeds). ΔMAE : relative reduction of $\text{MAE}(\nu_{12})$ of KAN versus linear (positive favours KAN), paired bootstrap 95% CI. Bold: CI excludes zero.

Mode	$R^2(\nu_{12})$ Linear / KAN		ΔMAE KAN vs. Linear	
	seed 123	seed 7	seed 123	seed 7
Single sample	0.852 / 0.904	0.815 / 0.911	+13.3% [−0.2, 24.4]	+ 18.2% [3.8, 30.4]
Single sample + refinement	0.994 / 0.997	0.998 / 0.999	+33.2% [−15.1, 60.5]	+ 33.9% [11.7, 52.3]
Best-of-30	0.977 / 0.974	0.982 / 0.980	− 25.2% [−56.2, −1.0]	− 19.9% [−42.1, −1.7]
Best-of-30 + refinement	0.998 / 0.997	0.998 / 0.999	+11.3% [−38.7, 43.6]	+ 44.1% [24.9, 59.7]

Table 7: Which method to use for auxetic inverse design, based on the FE-verified comparisons in this paper. “Generative”: cVAE best-of-30 with guarded binarization-aware refinement. “TO”: topology optimization from scratch, best of four starts.

Design situation	Use	Evidence
Many targets (thousands) inside the training range, anisotropy $r < 10$	Generative	Same ν_{12} accuracy with $\sim 8\times$ fewer FE solves; joint error 27% lower (pre-specified, CI 8–42%)
Few targets, or a single cell	TO	Cost of the training data is amortized only after $\sim 3\,500$ targets
Strong anisotropy, $r \geq 10$	TO	Generative pipeline much worse on 4 of 5 such targets (disconnected cells)
Manufacturability is a hard requirement	TO, or generative with $\lambda > 0$	Manufacturable 0.77–0.83 versus 0.32–0.35 at $\lambda = 0$, 0.72–0.77 at $\lambda = 0.1$ (joint-error advantage lost)
Unseen sign of ν , or magnitude beyond the data	TO	Generative beats retrieval only; magnitude saturates where data end
Any method	–	Optimize and select on the verified, binarized design; reject degenerate cells ($\min E_i < 10^{-3} E_0$)

against retrieval. (v) The cost advantage is amortized over about 3 500 targets. (vi) The aesthetic score is a heuristic that has not been validated against human judgement. (vii) The method of refinement was chosen after a failure on set A; the replication on sets B and C and the pre-specified test on set C mitigate, but do not remove, this dependence.

5 Conclusions

For inverse design of two-dimensional auxetic cells, a cVAE with best-of-30 sampling and latent refinement matches converged topology optimization in verified ν_{12} with about eight times fewer FE solves and achieves a lower joint error for moderately anisotropic targets, replicated on three disjoint target sets. The ingredient that makes this possible is to evaluate the refinement objective on the design that will be verified: binarized, resampled and checked for degeneracy. The generative pipeline is the method of choice when many in-range targets of moderate anisotropy must be served; topology optimization from scratch remains the better choice for strongly anisotropic targets, for targets outside the training data and when only a few cells are needed. Manufacturability can be bought back by penalizing it in the same refinement, at the cost of the accuracy advantage. The same principle applies to topology optimization itself: the verified design, not the projected field, is what has to be optimized and compared.

A Periodic-edge enforcement

The earlier pipeline replaced opposite boundary columns and rows by their mean before thresholding, both at verification and inside refinement, and counted edge mismatch in the manufacturability score. Because the FE mesh uses periodic boundary conditions on elements, every pixel cell tiles by translation

and this step is unnecessary. It also moved the verified properties of training cells away from their labels (MAE in ν_{12} 0.055 with the step, 0.032 without). On set A, removing it left the accuracy after best-of-30 and refinement unchanged. The decision to remove it was made before the numbers of this paper were recomputed; several results improved afterwards, which is why both versions are reported. Main changes (with $\rightarrow$ without): single-sample refinement ΔMAE +91.8% $\rightarrow$ +92.7%; best-of-30 refinement ΔMAE +62.6% $\rightarrow$ +68.3%; manufacturable after best-of-30 and guarded refinement 0.24 $\rightarrow$ 0.32 (with the edge criterion removed from the definition); magnitude sweep at $\nu_{12}^* = -2.0$, aware refinement, 12.5% $\rightarrow$ 1.2%; joint-error advantage at $r < 10$ on set C +27% $\rightarrow$ +35%, ν_{12} advantage +19% (CI includes zero) $\rightarrow$ +37%; Spearman mean of the composite score 0.693 $\rightarrow$ 0.718. The pre-specified results (set C stratified test, hybrid) are those obtained with the step.

Statements and declarations

Replication of results. Code (data generation, FE homogenization with exact sensitivities, cVAE training, refinement, topology-optimization baseline), trained checkpoints, the target sets and all per-target outputs underlying the tables and figures will be made available at [repository URL and Zenodo DOI to be inserted]. Every number in the paper is produced by scripts in docs/paper1/scripts from these outputs.

Funding. [To be completed.]

Competing interests. The author declares no competing interests. [To be confirmed.]

Use of generative AI. [To be completed by the author: AI assistance (Claude, Anthropic) was used for code development, data analysis and drafting of the manuscript; the author reviewed all content and takes full responsibility for it.]

Author contributions. [CRediT statement to be completed.]

References

- [1] Roderic Lakes. Foam structures with a negative Poisson’s ratio. *Science*, 235(4792):1038–1040, 1987. doi: 10.1126/science.235.4792.1038.
- [2] Kenneth E. Evans and Andrew Alderson. Auxetic materials: functional materials and structures from lateral thinking! *Advanced Materials*, 12(9):617–628, 2000. doi: 10.1002/(SICI)1521-4095(200005)12:9<617::AID-ADMA617>3.0.CO;2-3.
- [3] Katia Bertoldi, Vincenzo Vitelli, Johan Christensen, and Martin van Hecke. Flexible mechanical metamaterials. *Nature Reviews Materials*, 2:17066, 2017. doi: 10.1038/natrevmats.2017.66.
- [4] Ole Sigmund. Materials with prescribed constitutive parameters: an inverse homogenization problem. *International Journal of Solids and Structures*, 31(17):2313–2329, 1994. doi: 10.1016/0020-7683(94)90154-6.
- [5] Ulrik D. Larsen, Ole Sigmund, and Siebe Bouwstra. Design and fabrication of compliant micromechanisms and structures with negative Poisson’s ratio. *Journal of Microelectromechanical Systems*, 6(2):99–106, 1997. doi: 10.1109/84.585787.
- [6] Martin P. Bendsøe and Ole Sigmund. *Topology Optimization: Theory, Methods, and Applications*. Springer, Berlin, 2003. doi: 10.1007/978-3-662-05086-6.

- [7] Liwei Wang, Yu-Chin Chan, Faez Ahmed, Zhao Liu, Ping Zhu, and Wei Chen. Deep generative modeling for mechanistic-based learning and design of metamaterial systems. *Computer Methods in Applied Mechanics and Engineering*, 372:113377, 2020. doi: 10.1016/j.cma.2020.113377.
- [8] Yunwei Mao, Qi He, and Xuanhe Zhao. Designing complex architected materials with generative adversarial networks. *Science Advances*, 6(17):eaaz4169, 2020. doi: 10.1126/sciadv.aaz4169.
- [9] Siddhant Kumar, Stephanie Tan, Li Zheng, and Dennis M. Kochmann. Inverse-designed spinodoid metamaterials. *npj Computational Materials*, 6:73, 2020. doi: 10.1038/s41524-020-0341-6.
- [10] Jan-Hendrik Bastek, Siddhant Kumar, Bastian Telgen, Raphaël N. Glaesener, and Dennis M. Kochmann. Inverting the structure–property map of truss metamaterials by deep learning. *Proceedings of the National Academy of Sciences*, 119(1):e2111505119, 2022. doi: 10.1073/pnas.2111505119.
- [11] Hunter T. Kollmann, Diab W. Abueidda, Seid Koric, Erman Guleryuz, and Nahil A. Sobh. Deep learning for topology optimization of 2D metamaterials. *Materials & Design*, 196:109098, 2020. doi: 10.1016/j.matdes.2020.109098.
- [12] Xiaoyang Zheng, Ta-Te Chen, Xiaofeng Guo, Sadaki Samitsu, and Ikumu Watanabe. Controllable inverse design of auxetic metamaterials using deep learning. *Materials & Design*, 211:110178, 2021. doi: 10.1016/j.matdes.2021.110178.
- [13] Helda Pahlavani, Kostas Tsifoutis-Kazolis, Mauricio C. Saldivar, Prerak Mody, Jie Zhou, Mohammad J. Mirzaali, and Amir A. Zadpoor. Deep learning for size-agnostic inverse design of random-network 3D printed mechanical metamaterials. *Advanced Materials*, 36(6):2303481, 2024. doi: 10.1002/adma.202303481.
- [14] Jan-Hendrik Bastek and Dennis M. Kochmann. Inverse design of nonlinear mechanical metamaterials via video denoising diffusion models. *Nature Machine Intelligence*, 5:1466–1475, 2023. doi: 10.1038/s42256-023-00762-x.
- [15] Stephan Hoyer, Jascha Sohl-Dickstein, and Sam Greydanus. Neural reparameterization improves structural optimization, 2019. arXiv:1909.04240.
- [16] Aaditya Chandrasekhar and Krishnan Suresh. TOuNN: Topology optimization using neural networks. *Structural and Multidisciplinary Optimization*, 63:1135–1149, 2021. doi: 10.1007/s00158-020-02748-4.
- [17] Rebekka V. Woldseth, Niels Aage, J. Andreas Bærentzen, and Ole Sigmund. On the use of artificial neural networks in topology optimisation. *Structural and Multidisciplinary Optimization*, 65(10): 294, 2022. doi: 10.1007/s00158-022-03347-1.
- [18] Brandon Trabucco, Aviral Kumar, Xinyang Geng, and Sergey Levine. Conservative objective models for effective offline model-based optimization. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 10358–10368, 2021.
- [19] James K. Guest, Jean H. Prévost, and Ted Belytschko. Achieving minimum length scale in topology optimization using nodal design variables and projection functions. *International Journal for Numerical Methods in Engineering*, 61(2):238–254, 2004. doi: 10.1002/nme.1064.
- [20] Fengwen Wang, Boyan Stefanov Lazarov, and Ole Sigmund. On projection methods, convergence and robust formulations in topology optimization. *Structural and Multidisciplinary Optimization*, 43 (6):767–784, 2011. doi: 10.1007/s00158-010-0602-y.

- [21] Martin P. Bendsøe. Optimal shape design as a material distribution problem. *Structural Optimization*, 1:193–202, 1989. doi: 10.1007/BF01650949.
- [22] Ole Sigmund. A 99 line topology optimization code written in Matlab. *Structural and Multidisciplinary Optimization*, 21(2):120–127, 2001. doi: 10.1007/s001580050176.
- [23] Erik Andreassen, Anders Clausen, Mattias Schevenels, Boyan S. Lazarov, and Ole Sigmund. Efficient topology optimization in MATLAB using 88 lines of code. *Structural and Multidisciplinary Optimization*, 43(1):1–16, 2011. doi: 10.1007/s00158-010-0594-7.
- [24] Blaise Bourdin. Filters in topology optimization. *International Journal for Numerical Methods in Engineering*, 50(9):2143–2158, 2001. doi: 10.1002/nme.116.
- [25] J. M. Guedes and N. Kikuchi. Preprocessing and postprocessing for materials based on the homogenization method with adaptive finite element methods. *Computer Methods in Applied Mechanics and Engineering*, 83(2):143–198, 1990. doi: 10.1016/0045-7825(90)90148-F.
- [26] Erik Andreassen and Casper Schousboe Andreasen. How to determine composite material properties using numerical homogenization. *Computational Materials Science*, 83:488–495, 2014. doi: 10.1016/j.commatsci.2013.09.006.
- [27] Liang Xia and Piotr Breitkopf. Design of materials using topology optimization and energy-based homogenization approach in Matlab. *Structural and Multidisciplinary Optimization*, 52(6):1229–1241, 2015. doi: 10.1007/s00158-015-1294-0.
- [28] Krister Svanberg. The method of moving asymptotes—a new method for structural optimization. *International Journal for Numerical Methods in Engineering*, 24(2):359–373, 1987. doi: 10.1002/nme.1620240207.
- [29] M. D. McKay, R. J. Beckman, and W. J. Conover. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, 21(2):239–245, 1979. doi: 10.2307/1268522.
- [30] I. M. Sobol’. On the distribution of points in a cube and the approximate evaluation of integrals. *USSR Computational Mathematics and Mathematical Physics*, 7(4):86–112, 1967. doi: 10.1016/0041-5553(67)90144-9.
- [31] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Kopf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- [32] Tom Gustafsson and G. D. McBain. scikit-fem: A Python package for finite element assembly. *Journal of Open Source Software*, 5(52):2369, 2020. doi: 10.21105/joss.02369.
- [33] Diederik P. Kingma and Max Welling. Auto-encoding variational Bayes. In *International Conference on Learning Representations (ICLR)*, 2014. arXiv:1312.6114.
- [34] Kihyuk Sohn, Honglak Lee, and Xinchen Yan. Learning structured output representation using deep conditional generative models. In *Advances in Neural Information Processing Systems*, volume 28, 2015.
- [35] Samuel R. Bowman, Luke Vilnis, Oriol Vinyals, Andrew Dai, Rafal Jozefowicz, and Samy Bengio. Generating sentences from a continuous space. In *Proceedings of the 20th SIGNLL Conference on Computational Natural Language Learning*, pages 10–21, 2016. doi: 10.18653/v1/K16-1002.

- [36] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015. arXiv:1412.6980.
- [37] Dong C. Liu and Jorge Nocedal. On the limited memory BFGS method for large scale optimization. *Mathematical Programming*, 45:503–528, 1989. doi: 10.1007/BF01589116.
- [38] B. M. Lempriere. Poisson’s ratio in orthotropic materials. *AIAA Journal*, 6(11):2226–2227, 1968. doi: 10.2514/3.4974.
- [39] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993. doi: 10.1201/9780429246593.
- [40] Ziming Liu, Yixuan Wang, Sachin Vaidya, Fabian Ruehle, James Halverson, Marin Soljačić, Thomas Y. Hou, and Max Tegmark. KAN: Kolmogorov–Arnold networks, 2024. arXiv:2404.19756.
- [41] Vishwanath Saragadam, Daniel LeJeune, Jasper Tan, Guha Balakrishnan, Ashok Veeraraghavan, and Richard G. Baraniuk. WIRE: Wavelet implicit neural representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 18507–18516, 2023.